

Algorithmic Pseudocode Sample 33

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

Require: $\{\beta_i\}_{i=1}^N, \{\alpha_i\}_{i=1}^N, \text{csm}, \hat{\mathbf{y}}, \lambda_1, \lambda_2, r, N, M, \mathbf{M}_{\mathbf{u}}$. $\triangleright \text{csm} = \{\text{csm}_1, \dots, \text{csm}_n\}$, M_u is the undersampling mask

- 1: $\mathbf{x}_N \sim \mathcal{N}(\mathbf{F}^{-1}(\mathbf{M}_l \mathbf{y}), \mathcal{F}_h)$
- 2: **for** $i = N - 1$ to 0 **do**
- 3: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 4: $\mathbf{g} \leftarrow \mathcal{F}_h(\mathbf{s}_{\theta^*}(\mathbf{x}_{i+1}, i + 1))$
- 5: $\mathbf{G} = \sum_{j=1}^n \text{csm}_j^* \cdot \mathbf{F}^{-1}(\mathbf{F}(\text{csm} \cdot \mathbf{x}_i) \cdot \mathbf{M}_{\mathbf{u}} - \hat{\mathbf{y}})$
- 6: $\epsilon \leftarrow \lambda_1 (\|\mathbf{g}\|_2 / \|\mathbf{G}\|_2)$
- 7: $\mathbf{x}_i \leftarrow \mathbf{x}_{i+1} + \frac{1}{2} \beta_{i+1} \mathcal{F}_h(\mathbf{x}_i) + \beta_{i+1} (\mathbf{g} - \epsilon \mathbf{G}) + \sqrt{\beta_{i+1}} \mathcal{F}_h(\mathbf{z})$
- 8: **for** $k \leftarrow 1$ to M **do**
- 9: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 10: $\mathbf{g} \leftarrow \mathcal{F}_h(\mathbf{s}_{\theta^*}(\mathbf{x}_i^{k-1}, i))$
- 11: $\mathbf{G} = \sum_{j=1}^n \text{csm}_j^* \cdot \mathbf{F}^{-1}(\mathbf{F}(\text{csm} \cdot \mathbf{x}_i^k) \cdot \mathbf{M}_{\mathbf{u}} - \hat{\mathbf{y}})$
- 12: $\epsilon_1 \leftarrow 2\alpha_i (r \|\mathbf{z}\|_2 / \|\mathbf{g}\|_2)^2$
- 13: $\epsilon_2 \leftarrow \|\mathbf{g}\|_2 / (\lambda_2 \cdot \|\mathbf{G}\|_2)$
- 14: $\mathbf{x}_i^k \leftarrow \mathbf{x}_i^{k-1} + \epsilon_1 (\mathbf{g} - \epsilon_2 \mathbf{G}) + \sqrt{2\epsilon_1} \mathcal{F}_h(\mathbf{z})$
- 15: **end for**
- 16: $\mathbf{x}_{i-1}^0 \leftarrow \mathbf{x}_i^M$
- 17: **end for**
- return** $\mathbf{x}_0^0$

$\triangleright \mathbf{x}_0 = \mathbf{x}_i^0$